A Tripartite Modular Architecture for Adaptive Medical Dialogue Triage and Automated Clinical Documentation

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***Abstract*—Conversational AI assistants have the potential to automate a large proportion of the patient triage process. However, current AI models typically fall into two categories. One category consists of models that are single turn question answering models, whilst the other relies on having a general intent classifier and hard-coded responses without adapting to patient feedback. None of these models produce a record usable to Physicians. We present a three-stage modular framework for producing a clinical summary. First, a dialogue policy engine using sentence embeddings using the *all-mpnet-base-v2* model and class weighted LightGBM model selects one of 36 clinical actions to perform on the patient at a rate of 4.2 ms per action on a standard CPU. Second, the *LLaMA-3.1-8B-Instruct* model with QLoRA fine tuning converts the selected action into a single non-diagnostic question. Third, the finished interaction is converted into a clinical note with 15 fields using the fine-tuned *Flan-T5-small* model. On a benchmark dataset containing 1,000 consultations across 10 complaint types, our dialogue policy achieves 94.80% top-1 accuracy and 98.60% top-3 accuracy. Our summariser outputs an exact match for 5 of the fields and 99% for 2. The entire pipeline can fit within less than 6 GB VRAM.**

***Index Terms*—Medical chatbots, clinical dialogue systems, triage, clinical summarization, QLoRA, LightGBM, Flan-T5.**

I. Introduction

Symptom collection is often the first action performed in a clinical interaction and also one of the most repetitive actions. Building an automated history collector can reduce visit time, prevent some patients going to an emergency room by providing a baseline. However,Building such a system is difficult than it sounds, because it has to do three things at once.

1) Determine what part of the summary is missing (onset, pain, aggravating factors) and determine what to ask.
2) Avoid diagnostic reasoning and flag symptoms such as radiating chest pain or severe shortness of breath
3) Create a structured summary of the entire conversation.

Whilst existing models address some of these points, they fail to address all. Systems such as Retrieval Augmented ones such as Chen Chang's Medical Retrieval and Summarization (MRS) pipeline [1]. This uses dense retrieval and PEGASUS summarization with a frozen LLaMA-3 8B model to answer questions from the PubMedQA dataset. Whilst this is useful to gather information from the literature, each component is not consistent. There is no dialogue state, no questions asked about the previous answers and no patient history. Dialogue systems such as Kumar et al [2] do contain dialogue. They use a fine-tuned BERT model to determine the intent and extract entities. However, there are only 5 coarse intents in the intent space. The responses are templated and no summary is provided.

We are using a different way and split the problem into three specialised parts than asking one model to do everything. Our contributions are as follows.

- **Dialogue policy engine** Combines normalised MP- Net sentence embeddings, class weighted Light-GBM ensemble with 36 fine grained actions including an emergency `RED_FLAG` class, and runs in 4.2 ms on a CPU.
- A **response formulator**, a QLoRA-adapted LLaMA-3.1-8B-Instruct, that converts the selected action into one question.
- A **clinical summary generator**, a fine-tuned Flan-T5-small, that writes a 15-field EHR-style summary form whole conversation
- An evaluation on a 1,000-dialogue dataset with scenario-isolated splits, together with an honest account of where the summarizer still fails.

The rest of the paper is organised as follows. Section II reviews related work, Section III describes the architecture, Section IV covers the data and training setup, Section V reports results, and Section VI discusses limitations before Section VII concludes.

II. Related Work

A. Literature-Grounded Medical Question Answering

etrieval-augmented generation (RAG) gives an AI model to use every possible external way to answer questions . Chen and Chang [1] apply the idea to biomedicine: their MRS module retrieves with DPR, compresses passages with PEGASUS, and

uses a gating network before handing the result to LLaMA-3 8B, reaching 74.00% on PubMedQA. The task is reading comprehension over papers. It does not require tracking what a patient has already reported, deciding which question to ask or storing anything so it says little about intake.

B. Intent-Driven Medical Chatbots

Kumar et al. [2] target the interactive side. Their approach uses a fine-tuned BERT for intent and entity recognition, adds PII redaction, and sends replies through a Flask backend. They have a strong privacy work . The conversational part is lacking by a five-way intent space that cannot represent distinctions such as pain or fever magnitude by heuristic follow-up rules by the absence of any report or summary.

C. Controlling Language Models in Triage

General-purpose language models can process a large amount of medical text but knowledge is not all enough. Never failing case of an emergency Prompting a base model for this is unreliable: the zero-shot LLaMA-3-8B baseline used by Chen and Chang [1] breaks the single question rule in about a third of turns in our own evaluation (Section V). Thats the reason of our use of a small classifier in dialogue policy and use the LLM only for wording.

D. Efficient Adaptation and Summarization

Low-rank adaptation and its 4-bit variant, QLoRA, make it possible to specialise an 8B model on a single modest GPU. For summarization , instruction tuned encoder and decoder models such as Flan-T5 are small capable to fine-tune quickly and produce fixed format output reliably which all required in this situation.

E. Gap Summary

Table I compares the two closest systems with ours. Neither reference work offers multi turn policy learning over fine grained actions, a safety interrupt or structured documentation but our design is aimed at exactly those gaps.

III. Proposed Architecture

The whole system is a closed loop with three stages (Fig. 1). In each turn the policy engine reads the dialogue so far and selects an action; the formulator turns that action into a question; and when the patient ends the session, the summary generator writes the note.

A. Dialogue Policy Engine

At turn t the history $H_t = \{(u_1, r_1), \ldots, u_t\}$ is serialised into a speaker-tagged string, and encoded by a frozen MPNet sentence encoder into a unit-length 768-dimensional vector:

$$\mathbf{z}_t = \frac{\mathcal{E}_{\text{MPNet}}(S_t)}{\|\mathcal{E}_{\text{MPNet}}(S_t)\|_2}. \tag{1}$$

A multi-class LightGBM ensemble maps $\mathbf{z}_t$ to a distribution over the 36 action labels:

$$P(y = c \mid \mathbf{z}_t) = \frac{\exp\big(g_c(\mathbf{z}_t)\big)}{\sum_{k=1}^{36} \exp\big(g_k(\mathbf{z}_t)\big)}. \tag{2}$$

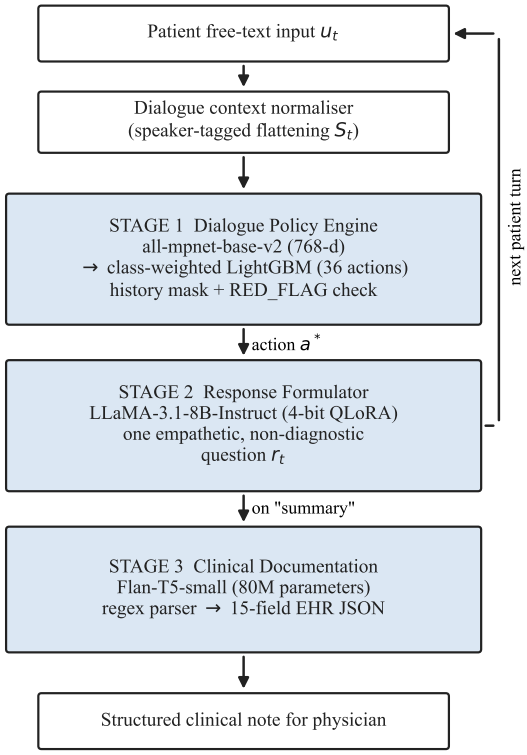


Fig. 1. Overall workflow. The policy engine selects an action a^*, the formulator verbalises it, and the loop repeats until the patient ends the session, at which point the summary generator produces the clinical note.

Some actions are far more common than others, so each training sample is weighted by the balanced inverse frequency

$$w_c = \frac{N}{C \cdot N_c}, \tag{3}$$

where N is the number of training examples, $C = 36$ and N_c is the count for action c.

Two rules sit on top of the classifier (Fig. 2). A history mask removes the question that have already been asked about, so the agent does not repeat itself. The emergency class is the exception: if the top-ranked prediction is `RED_FLAG`, the mask is ignored and the conversation goes straight to emergency routing. Because this check is a plain comparison on the classifier output, it can be audited and tested independently of the language model.

B. Response Formulator

Given the chosen action a^*, a LLaMA-3.1-8B-Instruct model writes the next turn. The base weights are stored in 4-bit NormalFloat with double quantisation, and low-rank matrices A and B are trained on the projection layers:

$$W_{\text{eff}} = W_{\text{frozen}} + \frac{\alpha}{r} BA, \qquad r = 16,\ \alpha = 32. \tag{4}$$

Generation is conditioned on the action, the history and a safety prompt:

$$r_t \sim P_\theta\big(r_t \mid a^*, H_t, \text{Prompt}_{\text{safety}}\big). \tag{5}$$

TABLE I
ARCHITECTURAL COMPARISON WITH THE CLOSEST REFERENCE WORKS

Dimension	Chen & Chang [1]	Kumar et al. [2]	Proposed system
Primary objective	Biomedical QA on scientific papers	Symptom intent classification	Adaptive multi-turn triage and documentation
Interaction mode	Single-turn literature QA	Multi-turn, rule-guided	Multi-turn, policy-guided
Policy / state tracking	None	BERT, 5 coarse intents	MPNet + LightGBM, 36 granular actions
Response generation	Prompted LLaMA-3 8B (zero-shot)	Static templates	LLaMA-3.1-8B, 4-bit QLoRA
Clinical documentation	None (short answer)	None (chat text only)	15-field EHR note (Flan-T5)
Emergency escalation	Absent	Absent	Deterministic `RED_FLAG` interrupt
Hardware	High-end GPU ($\geq$ 24 GB VRAM)	Cloud / CPU server, non-generative	Single workstation ($<$ 6 GB VRAM)

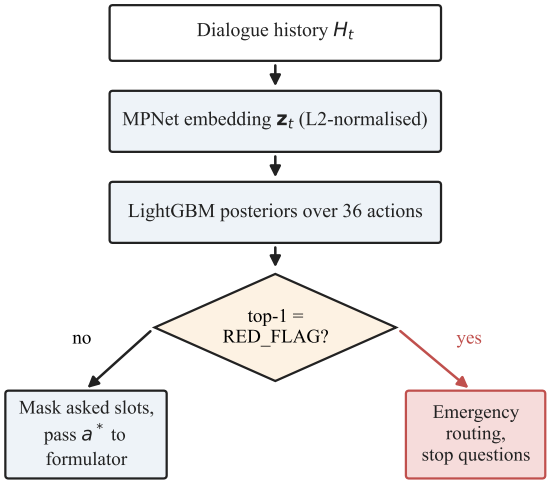


Fig. 2. Decision logic of the policy engine. The red-flag branch bypasses the history mask and the formulator.

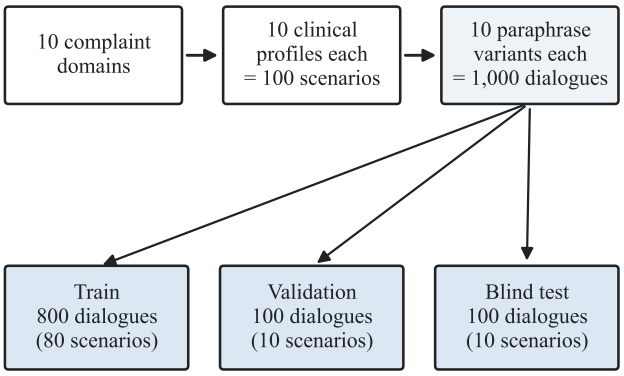


Fig. 3. Construction and splitting of the synthetic benchmark.

The prompt asks the model to (i) pose exactly one concise question about a^*, (ii) acknowledge what the patient has already said, (iii) avoid any diagnostic statement, and (iv) stop asking questions and give emergency instructions when $a^* = \texttt{RED_FLAG}$.

C. Clinical Summary Generator

When the session ends, the full transcript H_T goes to a Flan-T5-small model (80M parameters) fine-tuned with the usual autoregressive cross-entropy objective,

$$\mathcal{L}_{\text{Doc}} = -\sum_{i=1}^{M} \log P\big(y_i \mid y_{<i}, H_T; \theta_{\text{T5}}\big). \tag{6}$$

The target is a text rendering of 15 fields: chief complaint, onset, duration, location, severity, complaint characteristics, associated symptoms, negative findings, medical history, medications, allergies, red flags, triggers, relieving factors and social context. A deterministic regular-expression parser converts this text into a JSON record and an indented note for physician review.

IV. EXPERIMENTAL SETUP

A. Benchmark

We used roboflow along with some dataset and also some synthetic benchmark (Fig. 3) . The benchmark is based on ten main complaints, such as headache, low back pain, cough, abdominal pain, urinary symptoms, sore throat, knee pain, skin rash, dizziness and chest pain. Ten different clinical cases for each of these complaints were developed. In total there were 100 scenarios, each of which was described using ten para phrases to get 1,000 consultations of 14–22 turns, respectively. Each dialogue had a reference 15 field summary..

Splits are made at the scenario level: 800 training dialogues (80 scenarios), 100 validation dialogues (10 scenarios) and 100 blind-test dialogues (10 scenarios). No scenario, and therefore no paraphrase of one, appears in more than one split.

B. Training Configuration

Table II lists the settings for the three components.

V. RESULTS

A. Action Selection

We compared the policy engine with a rule matcher, BERT-base and BioBERT intent classifiers, RoBERTa-large, and

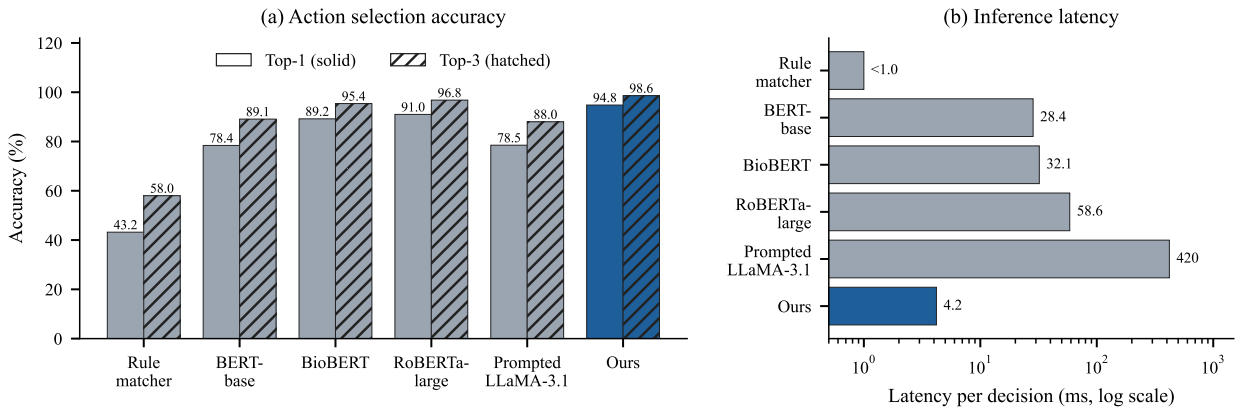


Fig. 4. Policy engine comparison. (a) Top-1 and top-3 action accuracy. (b) Per-decision latency on a log scale; the rule matcher is reported as $<$1 ms.

TABLE II
TRAINING CONFIGURATION

Component	Settings
Policy engine	*all-mpnet-base-v2* embeddings; LightGBM multi-class, 150 trees, max depth 6, learning rate 0.05, class weighting on
Response formulator	Llama-3.1-8B-Instruct, NF4 (bitsandbytes); LoRA r=16, α=32, dropout 0.05; batch size 2, gradient accumulation 4; AdamW, cosine schedule from 2×10^{-4}; 3 epochs
Summary generator	*flan-t5-small*; Hugging Face `Seq2SeqTrainer`; source 512 / target 256 tokens; learning rate 5×10^{-4}; weight decay 0.01; beam width 4 at inference

TABLE III
ACTION CLASSIFICATION ON THE BLIND TEST SET

Model	Top-1	Top-3	$F1_{macro}$	Latency
Rule matcher [2]	43.20%	58.00%	0.39	$<$1.0 ms
BERT-base [2]	78.40%	89.10%	0.74	28.4 ms
BioBERT-v1.1	89.20%	95.40%	0.86	32.1 ms
RoBERTa-large	91.00%	96.80%	0.88	58.6 ms
Prompted LLaMA-3.1-8B	78.50%	88.00%	0.75	420.0 ms
MPNet + LightGBM (ours)	**94.80%**	**98.60%**	**0.93**	**4.2 ms**

a prompted LLaMA-3.1-8B, all on the same 36-way task (Table III, Fig. 4).

The MPNet and LightGBM combination is the most accurate of all six systems, at 94.80% top-1 and 98.60% top-3, and it is 3.8 points ahead of RoBERTa-large at top-1. The more practical difference is speed. At 4.2 ms on a CPU, it is roughly 14 times faster than RoBERTa-large (58.6 ms) and about 100 times faster than the prompted LLaMA model (420 ms), which also scores 16 points lower. Only the rule matcher is faster and it gets less than half of the actions right. We used this as a deciding factor for a closed set of clinical actions, a good frozen embedding plus a tree ensemble is enough, and that spending an LLM call on policy selection is unnecessary.

B. Response Quality

Quality of responses was evaluated based on 200 interaction turns. Medical experts judged empathic responses on a scale from 1 to 5, while we calculated the number of turns with exactly one question asked, hallucinated diagnostic assertions, and repetitions of an earlier question (Fig. 5).

The fine-tuned LLaMA-3.1-8B scored 4.82 in empathy, adhering to the one-question principle in 98.4% of interaction turns, making unsupported diagnosis assertions in 0.2%, and repeating an earlier question in 1.1%. Templates are completely compliant and never hallucinate, but their score is 2.10 in empathy and they repeat in 24.5% of interaction turns. The baseline LLaMA-3 model, on the other hand, is natural-sounding (3.60) and violates the one-question principle in almost a third of turns, but it hallucinates in 14.8%. Thus, the fine-tuned model is the only one that performs well on all four criteria simultaneously, and it should be used for wording after the policy engine has decided what to ask.

C. Clinical Documentation

Field-level accuracy was computed by exact match on the 100 blind-test transcripts (Fig. 6).

Five fields (chief complaint, onset, location, symptom characteristics and past medical history) are exact in all 100 cases, and duration and severity are correct in 99. Relieving factors reach 96%, and negative findings, associated symptoms, medications and triggers fall between 89% and 92%. All 100 outputs were parseable, so the regex stage produced no schema errors. Three fields are weak: social context (50%), allergies (10%) and red flags (1%).

VI. DISCUSSION AND LIMITATIONS

The low scores on allergies and red flags tell an important story. In 94% of our test dialogues, the right answer for these fields was a simple "no"—like "no known allergies" or "no red flags". But our evaluation tool only accepts exact matches. Any different wording gets marked wrong. So the low numbers don't mean the system missed the information; they mostly mean our metric was too strict.

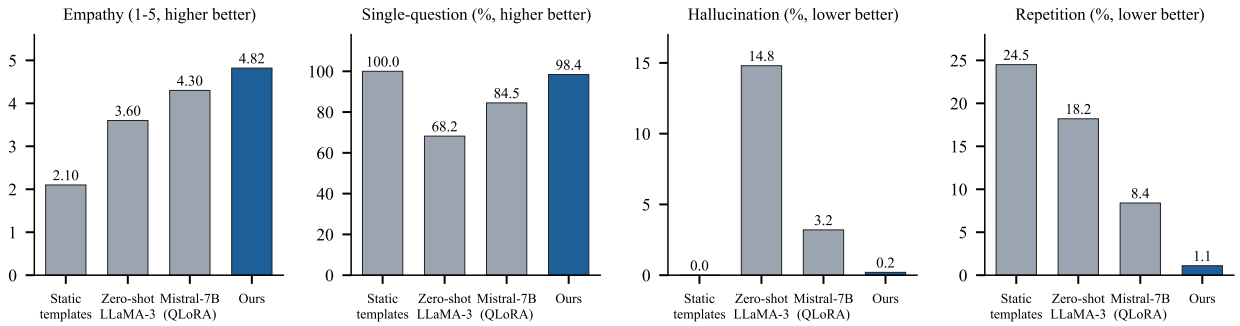


Fig. 5. Response formulation across four generators. Higher is better for empathy and single-question compliance; lower is better for hallucination and repetition.

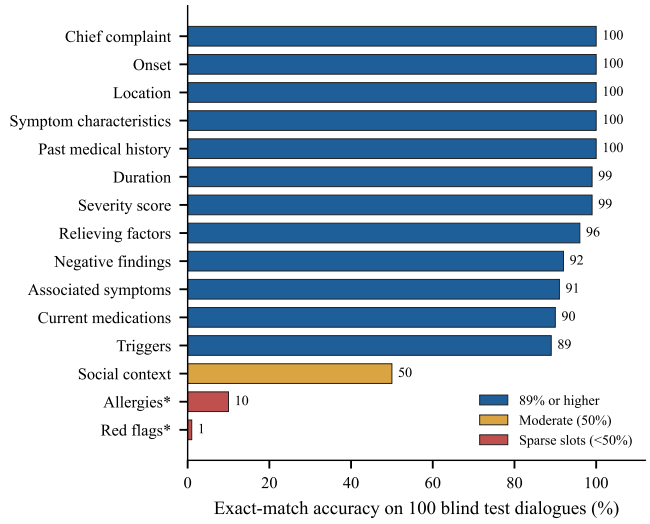


Fig. 6. Exact-match accuracy of the summary generator for each of the 15 fields. Fields marked with an asterisk are mostly empty in the test set and are discussed in Section VI.

We haven't checked this with a doctor yet or used a more flexible scoring method, so this is our best guess, not proven.

Social context is genuinely harder. The information is often implied rather than directly stated, so it's harder for the model to pick up.

There's a bigger limitation too: our test data is synthetic. We created it ourselves, and we only tested on 10 different scenarios. Scores of 100% on many fields suggest our dialogues are more orderly than real patient conversations. Real patients interrupt, give vague answers, and switch languages. We expect those to lower the scores. We don't claim these results would work in a real clinic.

The emergency rule is another weak spot. We tested it on dialogues that mostly were not emergencies. We never actually tested whether it catches real emergencies.

On empathy, medical experts only rated 200 conversation turns, which is a small sample. In future work, we should measure how well different raters agree on empathy scores.

Finally, we tested our question-writer against a prompted LLaMA model but not a simpler baseline. So we can't prove that our fancier approach is actually better—we just know it parses without errors.

VII. Conclusion

We are using a modular system for medical intake which uses different models for each parts as , what question to ask next and what will be the wording of that quetsion , for generation of final summary. The first part is a lightweight MPNet model combined with LightGBM. It picks from 36 clinical actions and gets it right 94.80% of the time in just 4.2 milliseconds on a regular CPU. The second part is a LLaMA-3.1-8B model that we fine-tuned with QLoRA. It writes one empathetic question at a time—never stating a diagnosis. The third part is a small Flan-T5 model that reads the whole conversation and writes a 15-field clinical note. It gets the core history fields exactly right. We put the emergency check inside the first part, the classifier. This makes it easy to understand and review what the system is doing. The most important work ahead is clear: test this on real patient conversations or ones written by doctors, use a fairer way to score negative findings (like "no allergies"), and properly test whether the emergency detection actually works.

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